A Counterexample to the Unnormalized One-Homogeneous Weighted Minkowski Extension

Run `fa_banach_001`, lane 16

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Status

Claim status: candidate counterexample, likely valid. This packet concerns Remark 2.25 of Kryvonos–Langharst [1], which states that when the measure is 1-homogeneous the weighted surface area measure is 0-homogeneous, so the scaling argument used in Lemma 2.24 fails and the case remains open.

The counterexample below shows that the natural unnormalized statement is false already in dimension two. It does not contradict the normalized existence theorem in the source paper, where a scalar factor $c_{\mu,K}$ is allowed.

Source Excerpt

The relevant open signal is Remark 2.25 on page 25 of [1].

$$d\nu(u) = A^{\alpha-1} dS^{\mu}_{\tilde{K}}(u) = dS^{\mu}_{A\tilde{K}}(u) = dS^{\mu}_{K}(u),$$

as claimed. □

Remark 2.25. For a measure that is 1-homogeneous, the surface area measure $dS^{\mu}_{L}(u)$ is 0-homogeneous; therefore, the above schema fails, and this case remains open.

Combining existence and uniqueness, we can extend the result of Livshyts [61]:

Theorem 2.26. *Let a Radon measure μ be even, p-concave, and α-homogeneous so that $\alpha \neq 1$ and either $\alpha > n$, $p > 0$ or $\alpha < n$ but $p = \frac{1}{}$. Suppose ν is*

Transcribed Open Signal

After proving that, for an α-homogeneous measure with $\alpha \neq 1$, the normalizing constant in the weighted Minkowski equation can be removed by dilating the body, the source paper states:

> For a measure that is 1-homogeneous, the surface area measure $dS^{\mu}_{L}(u)$ is 0-homogeneous; therefore, the above schema fails, and this case remains open.

Definitions

For a convex body $K \subset \mathbb{R}^2$ containing the origin in its interior, let S_K^μ denote the weighted surface area measure associated with a measure μ with density ϕ:

$$\int_{\mathbb{S}^1} f(u)\, dS_K^\mu(u) = \int_{\partial K} f(n_K(x))\phi(x)\, d\mathcal{H}^1(x).$$

We identify $u \in \mathbb{S}^1$ with $u(\theta) = (\cos\theta, \sin\theta)$, and write $d\theta$ for arc length measure on $\mathbb{S}^1$.

Proof Intuition

For the radial 1-homogeneous measure $d\mu(x) = |x|^{-1}\, dx$, the weighted surface area density of a smooth planar body is the curvature radius divided by the distance of the boundary point from the origin. Asking for this density to be a constant a imposes the differential equation

$$h + h'' = a\sqrt{h^2 + (h')^2}$$

on the support function. After the substitution $y = h'/h$, the equation becomes a one-dimensional autonomous ODE. Such an ODE has no nonconstant periodic trajectories, while a support function must be periodic. The only constant periodic possibility is the Euclidean circle, and it corresponds to $a = 1$. Therefore the uniform data $2\, d\theta$ cannot occur.

Theorem 1. *Let μ be the even Radon measure on $\mathbb{R}^2$ with density $\phi(x) = |x|^{-1}$. Then μ is 1-homogeneous. For every constant $a > 0$, if a symmetric convex body $K \subset \mathbb{R}^2$ with $0 \in \operatorname{int} K$ satisfies*

$$dS_K^\mu = a\ d\theta \qquad \text{on } \mathbb{S}^1,$$

then $a = 1$. In particular, for the finite even measure $\nu = 2\, d\theta$, which is not concentrated on any hemisphere, there is no symmetric convex body K such that $dS_K^\mu = d\nu$.

Proof. The density $|x|^{-1}$ is locally integrable in $\mathbb{R}^2$, even, and smooth away from the origin. Moreover,

$$\mu(rA) = \int_{rA} |x|^{-1}\, dx = r\int_A |y|^{-1}\, dy = r\mu(A),$$

so μ is 1-homogeneous.

Assume that K satisfies $dS_K^\mu = a\ d\theta$. Let $h(\theta) = h_K(u(\theta))$ be its support function. Since $0 \in \operatorname{int} K$, h is bounded below by a positive constant. We use the standard planar support-function facts

$$dS_K = (h'' + h)\, d\theta$$

in the distributional sense, and, whenever S_K is absolutely continuous, the boundary point with outer normal $u(\theta)$ is

$$x(\theta) = h(\theta)u(\theta) + h'(\theta)u'(\theta) \quad \text{for a.e. } \theta.$$

Thus $|x(\theta)| = \sqrt{h(\theta)^2 + h'(\theta)^2}$.

Because K contains the origin in its interior, $|x|$ is bounded above and below on ∂K. Hence the relation $dS_K^\mu = |x(\theta)|^{-1}\, dS_K = a\ d\theta$ implies that S_K is absolutely continuous and $h \in W^{2,1}_{\text{loc}}$. Its density satisfies, for a.e. θ,

$$h + h'' = a\sqrt{h^2 + (h')^2}. \tag{1}$$

Set $y = h'/h$. Then y is absolutely continuous and, using (1),

$$\begin{aligned} y' &= \frac{h''h - (h')^2}{h^2} \\ &= a\sqrt{1+y^2} - 1 - y^2 =: F_a(y) \end{aligned}$$

for a.e. θ. Since F_a is smooth, y is in fact C^1 and solves the autonomous ODE $y' = F_a(y)$. Also h is periodic, so y is periodic.

A one-dimensional autonomous ODE has no nonconstant periodic solutions: if a periodic C^1 solution attains a maximum or minimum at a non-equilibrium value, its derivative there is nonzero, while if it attains an equilibrium value then uniqueness forces the solution to be constant. Hence $y \equiv c$ for some constant c.

The equation $h' = ch$ and periodicity of h force $c = 0$. Substituting $y = 0$ into $y' = F_a(y)$ gives $0 = F_a(0) = a - 1$. Therefore $a = 1$. Taking $a = 2$ gives the announced counterexample. $\square$

Verification Notes

The proof uses only standard planar convex-body support-function calculus and the elementary phase-line obstruction for one-dimensional autonomous ODEs. The argument rules out all convex bodies, not only smooth bodies: absolute continuity of S_K^μ and the bounded positive weight $|x|^{-1}$ on ∂K force the ordinary surface area measure S_K to be absolutely continuous, so the support-function ODE holds in $W^{2,1}$.

The included script in `code/` prints the equilibria of $F_a(y) = a\sqrt{1+y^2} - 1 - y^2$ and checks that the only constant compatible with a periodic support function is $a = 1$.

Novelty and Search Bounds

Run-index search was performed for `2111.10923`, the paper title, `weighted Minkowski`, `projection measures`, and close Minkowski/projection-body phrases in the run's lightweight registry, solution, attempt, and proof-gap indexes; no direct duplicate was found. Web searches for close phrases around `1-homogeneous`, `weighted surface area measure`, `remains open`, and the paper title surfaced the source arXiv record but no later explicit answer.

Limitations

This packet disproves the unnormalized alpha-one extension suggested by Remark 2.25. The source paper's normalized theorem with a scalar $c_{\mu,K}$ remains untouched: for example, the Euclidean disk has $dS_K^\mu = d\theta$, so $2\,d\theta = 2\,dS_K^\mu$ is representable when a scalar factor is allowed.

Human Review Recommendation

Check the passage from $dS_K^\mu = a\,d\theta$ to the planar support-function ODE for nonsmooth bodies. If that standard measure-theoretic step is accepted, the rest of the proof is elementary and the counterexample should be promoted.

References

[1] Liudmyla Kryvonos and Dylan Langharst, *Weighted Minkowski's Existence Theorem and Projection Bodies*, arXiv:2111.10923, 2021.

[2] Rolf Schneider, *Convex Bodies: The Brunn–Minkowski Theory*, second expanded edition, Encyclopedia of Mathematics and its Applications 151, Cambridge University Press, 2014.